

Haile Lakew

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Senior Fullstack Developer with 8 years of experience specializing in high-traffic e-commerce storefronts and scalable React/Node.js solutions. Expert in **GitHub Actions CI/CD automation**, **custom Webpack builds**, and **web performance optimization** that drive customer engagement. Dedicated mentor and technical lead focused on bridging the gap between engineering and business goals.

Work Experience

Senior Software Engineer The Home Depot (Jan 2018 - Mar 2026)

Revenue Growth & Business Impact

- Architected high-impact Homepage features that generated \$100M+ in revenue (81bps YoY boost), and improved bounce rate by 200bps during peak Black Friday traffic
- Engineered performance optimizations that scaled systems to support 9.2M daily visits, driving a 28% increase in event revenue and a record \$336M in total campaign sales
- Led **A/B experimentation** for Homepage Hero components, increasing click-through volume (1.3M to 2.2M), and generated \$82.M in incremental sales; successfully driving its full transition from proof of concept to a permanent, scalable architecture
- Developed a self-service web application that empowered business partners to deploy event-specific modules via a custom **Content Management System (CMS)**; this directly enabled \$304M in attributed revenue (\$294M from Product Description Page Badges and \$10.2M from Digital Endcaps) across 57M visits

Technical Architecture & Scalability

- Led a Header refactor to use **Apollo GraphQL** and a headless CMS (Contentful), improving findability by 22% and scaling the system to support 155M+ annual interactions
- Built systems using custom **Webpack** and **GitHub Actions**, reducing **CI/CD** build times by 83% (30 min to 5 min); engineered **Node.js CSS injection** scripts to decouple styles and **prerender HTML**, resulting in a ~20% reduction in bundle size

Engineering Efficiency & DevOps

- Ensured 99.9% site reliability across high traffic systems at an enterprise scale, utilizing **New Relic** and **Blue Triangle** to monitor **Core Web Vitals (CWV)**; owned and managed release rollout through our cloud infrastructure on **Google Cloud Platform (GCP)**
- Initiated internal tooling and GitHub Actions to automate auditing, bundle size impact, and code quality enforcements; significantly accelerated delivery of high-impact features
- Early adopter of **Github Copilot** (2020 Developer Access) and **Agentic Software Development**; championed enterprise adoption by implementing **Model Context Protocol (MCP)** servers to extend AI agent capabilities, accelerating technical documentation onboarding and automated unit testing using **Jest** and **React Testing Library (RTL)**

Strategic Leadership & Mentorship

- Championed engineering excellence by mentoring mid-level and junior developers on design patterns and clean code standards; established a suite of developer tools (**Bash/Python**) that increased team velocity and enabled engineers to own large-scale projects. Mentored developers who successfully advance to Senior Engineer roles
- Created technical roadmaps for core storefront applications, translating business goals into actionable Epics and stories; leveraged **Jira** in collaboration with UX and Product to manage velocity and provide high-accuracy delivery estimates.

Projects

Open-Source & Creative Engineering

- 3D Optimization & Asset Pipeline**: Developed an optimization workflow for web based 3D environments, utilizing **Blender (shadow baking/UV unwrapping)** and implementing **Draco compression** and **WebP** to reduce 3D asset overhead; successfully reducing a 340MB asset to 1.3MB (99% reduction), decreasing **TTI (Time to Interactive)** and memory load
- WebGL & Shader Development**: Engineered a series of high-performance shader demos utilizing **GLSL** and **OpenGL**; implemented **FBO particles** and **Perlin Noise** algorithms to simulate complex fluid dynamics with peak **Core Web Vitals** and **60fps performance**
- Browser-Based Physics Engine**: Built 2D/3D platformers featuring custom physics constraints and real-time user input, focusing on **low-latency browser performance**

Technical Skills

Languages

Javascript (ES6+) / Typescript
HTML5 / CSS3 (SCSS / SASS)
Bash
Python
GLSL / OpenGL
Java

Frontend & Graphics

React / React Native
Next.js
Tailwind CSS
Framer Motion
Three.js / React Three Drei
Electron

Backend Architecture & APIs

Apollo GraphQL
REST
Headless CMS
Node.js
Springboot Java

Cloud & DevOps

Google Cloud Platform (GCP)
GitHub Actions (CI/CD)
Webpack
Vercel

AI & Efficiency

Model Context Protocol
Agentic Development
Large Language Models
Github Copilot

Performance & Monitoring

New Relic
Blue Triangle
Jest
React Testing Library (RTL)

Education

University of Texas at Dallas
B.S. COMPUTER SCIENCE
Cum Laude, 2018

Miscellaneous

Kali Linux Pen Testing Labs
Flipper Zero Programming
Raspberry Pi Projects
Classic Games Emulation